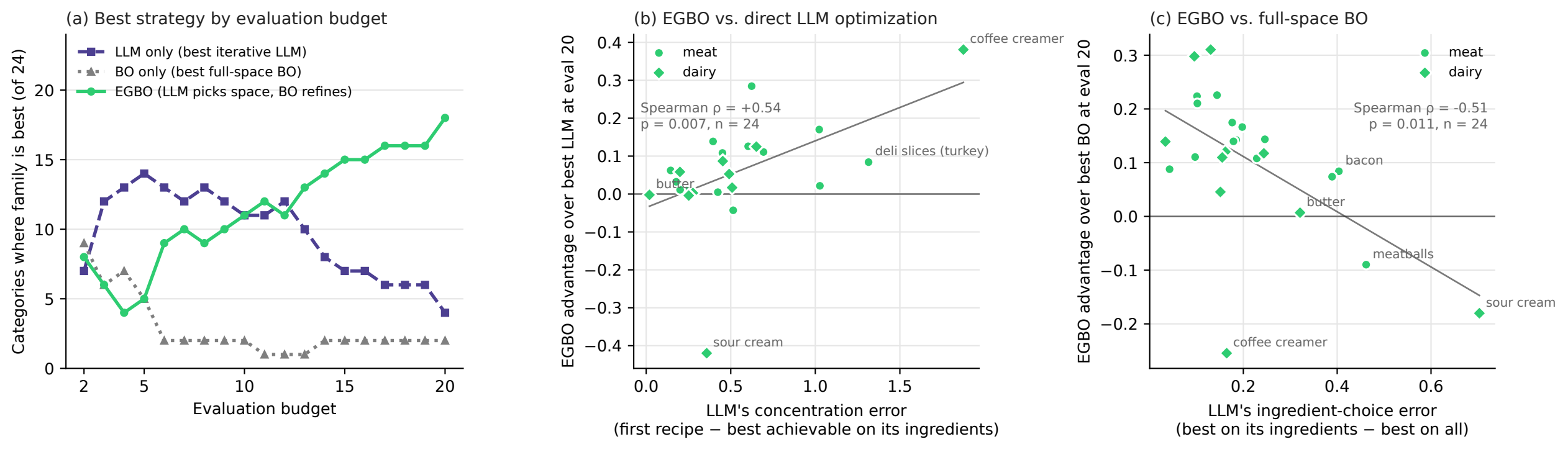
(a) Best strategy by evaluation budget
LLM only (best iterative LLM)
BO only (best full-space BO)
EGBO (LLM picks space, BO refines)
Categories where family is best (of 24)
Evaluation budget
(b) EGBO vs. direct LLM optimization
meat
dairy
Spearman ρ = +0.54
p = 0.007, n = 24
coffee creamer
deli slices (turkey)
butter
sour cream
EGBO advantage over best LLM at eval 20
LLM's concentration error
(first recipe − best achievable on its ingredients)
(c) EGBO vs. full-space BO
meat
dairy
Spearman ρ = -0.51
p = 0.011, n = 24
bacon
butter
meatballs
sour cream
coffee creamer
EGBO advantage over best BO at eval 20
LLM's ingredient-choice error
(best on its ingredients − best on all)